

1. Business Context and Problem Statement

MovieLens is an online database of film, television, and streaming content, offering cast, crew, trivia, and rating information alongside fan and critical reviews. Each year, in collaboration with a guest curator, MovieLens publishes annual insights around a chosen theme.

As the Data Analyst on this project, the objective was to build a Power BI dashboard analyzing classic movies, user demographics, and the relationship between viewer characteristics and movie ratings, in order to surface patterns that support the campaign's storytelling and audience-growth goals.

2. Data Modeling, Cleaning and Transformation

Three source files were used: movie.csv (movie_id, movie_name, year, genre), user.csv (user_id, age, gender, occupation, zipcode), and rating.csv (user_id, movie_id, rating, timestamp).

2.1 Data Model

A star schema was built with ratings as the central fact table, related to two dimension tables:

- users (1) → ratings (*) on user_id — single-direction filter
- movies (1) → ratings (*) on movie_id — single-direction filter
- movies (1) ↔ movie_genre_split (*) on movie_id — bidirectional, to allow genre-level filtering of ratings through the pipe-separated genre field

2.2 Cleaning and Transformation Steps

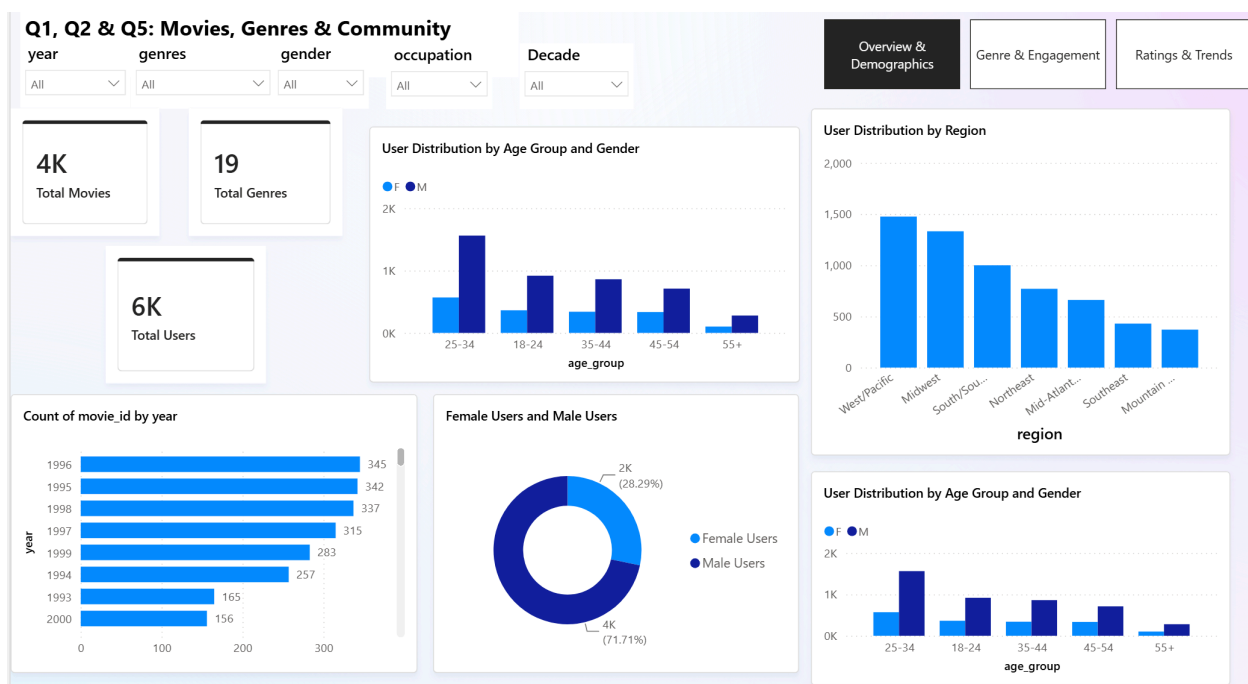
- Filtered movies to year ≤ 2000 to scope the dataset to classic films
- Split the pipe-delimited genre field into individual rows (movie_genre_split) for genre-level analysis, while retaining a one-row-per-movie movies table for movie counts
- Created a Genre Group custom column, consolidating individual/combination genre tags (e.g., Action, Action|Adventure) into a single category such as “Action based” for cleaner, more interpretable visuals

- Created a Decade column bucketing release years into decades (pre-1920 folded into the 1920s bucket) to support decade-level slicing
- Converted zipcode to text to preserve leading zeros; corrected the year field's data type from text to whole number after identifying and removing a duplicated header row that had been read in as data
- Imputed ages below 18 with the median age of the user's occupation (computed from an independent, non-circular query referencing the raw source file), then bucketed all users into five age groups: 18-24, 25-34, 35-44, 45-54, 55+
- Converted the rating timestamp from Unix epoch to a readable datetime and extracted a rating year for trend analysis
- Derived a region field from the leading digit of each user's zipcode as a substitute for a full geographic map, since a zip-to-state lookup table was not available within the provided dataset

3. Dashboard Overview

The report is organized across three pages, each with dedicated slicers (Year, Genre, Gender, Occupation, Decade, and Age Group as relevant) and synchronized filtering, plus a consistent navigation bar allowing movement between pages.

4. Q1, Q2 & Q5— Movies, Genres, and Community Overview



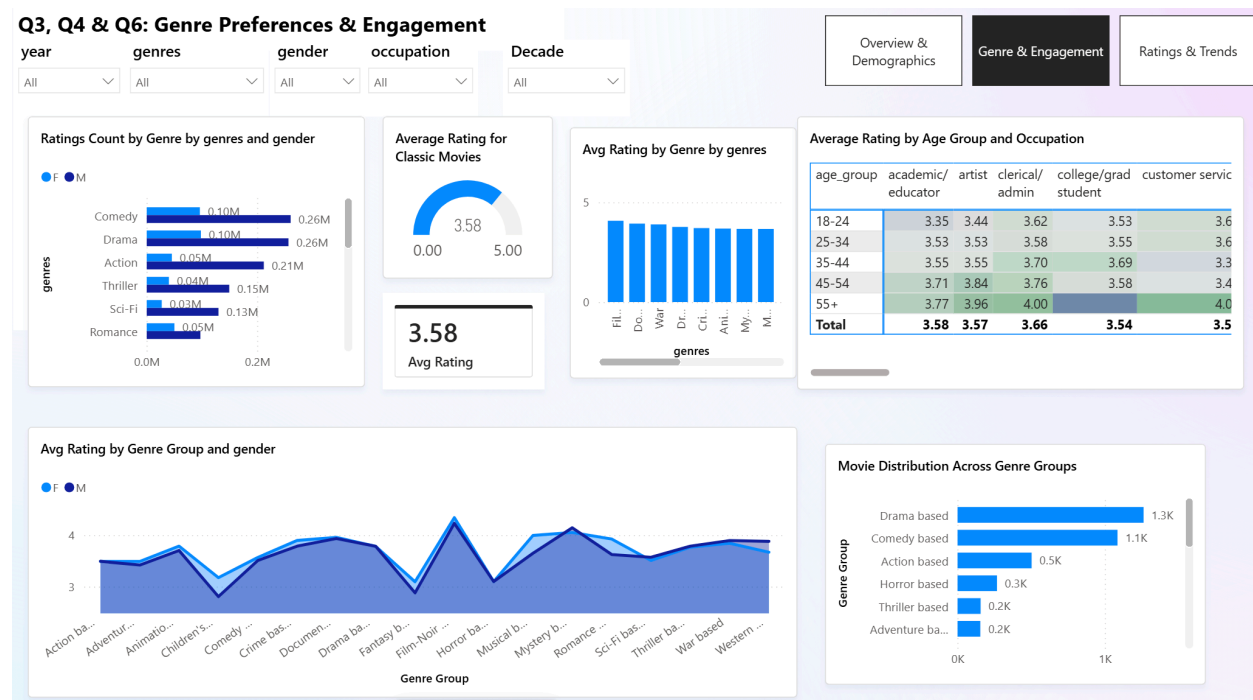
Q1: Movies and Genres Through 2000

The dataset contains 4,000 classic movies (released through 2000) spanning 19 distinct genres. Movie output by year shows a clear upward trend through the mid-to-late 1990s, with 1996 and 1995 representing the highest release volumes in the dataset, suggesting the platform's classic catalog is weighted toward more recent “classic” releases rather than older archival films.

Q2: MovieLens Community and Gender Diversity

The community comprises 6,000 users. Gender diversity skews notably male: 71.71% of users are male versus 28.29% female. This imbalance is a key consideration for the Movie Talkies: Classic campaign — marketing aimed at broadening the female viewer base could meaningfully diversify engagement.

5. Q3, Q4 & Q6 — Genre Preferences and Engagement



Q3: Genre Preferences by Gender

Comedy and Drama dominate rating volume across both genders, followed by Action and Thriller. When average rating (rather than volume) is examined by Genre Group, ratings stay

fairly consistent across groups (mostly in the 3.0–4.0 range) with mild peaks around Horror-based and Musical-based groups, indicating genre popularity (volume) and genre quality perception (rating) are not always aligned.

Q4: Overall Reception and Genre Distribution

The average rating across all classic movies is 3.58 out of 5, indicating generally positive reception. Movie counts are heavily concentrated in Drama-based ($\approx 1.3\text{K}$) and Comedy-based ($\approx 1.1\text{K}$) genre groups, while groups such as Documentary-based, Adventure-based, and Thriller-based have comparatively few titles — useful information for catalog-expansion planning.

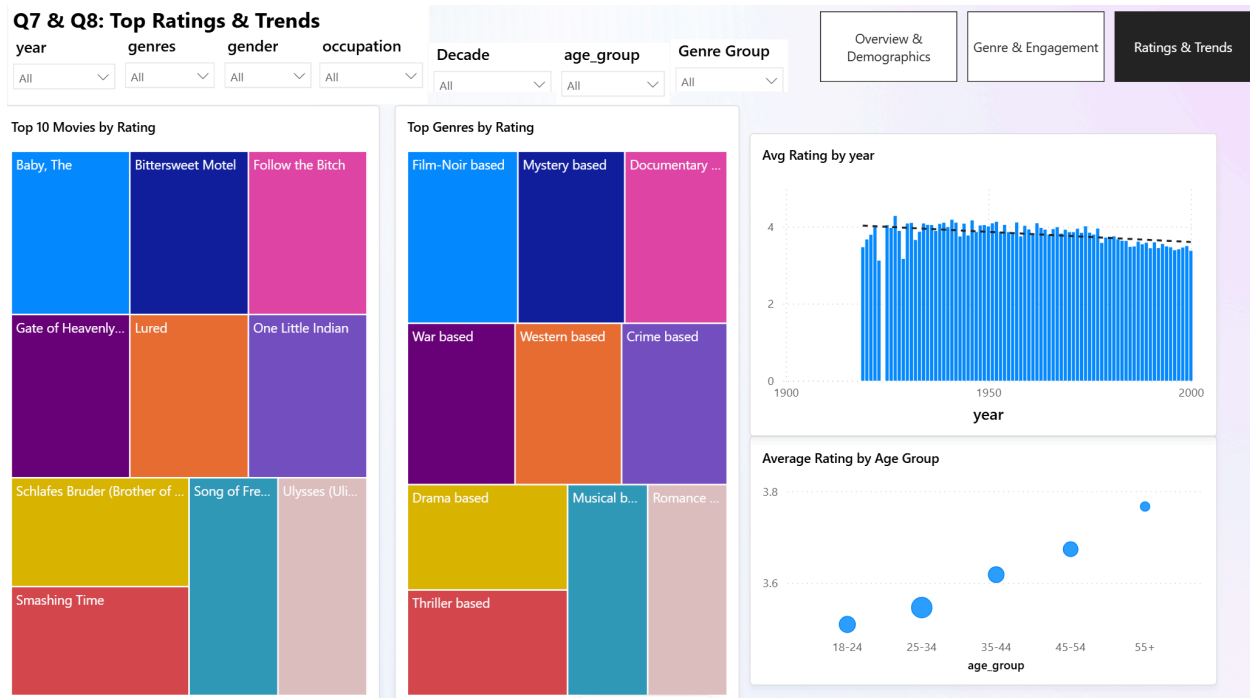
Q6: Engagement by Age Group and Occupation

The age-group $\times$ occupation matrix shows that older users (55+) in academic/educator and customer-service roles tend to rate movies most favorably (up to 4.05), while the youngest group (18-24) rates more moderately (typically 3.3–3.6). This suggests older, more established occupational segments are more receptive audiences for classic-movie content — a useful signal for targeted promotion.

6. Q5 — Demographic Distribution and Geography

(See Page 1 above.) Age-group and gender distribution shows the 25-34 segment as the largest, skewing male across every age bracket. Geographically, since a zip-to-state lookup was not available in the source data, users were grouped into regions using the leading digit of their zip code; the West/Pacific and Midwest regions show the highest concentration of users, with Mountain and Southeast regions comparatively underrepresented.

7. Q7 & Q8 — Top Ratings and Trends



Page 3: Ratings & Trends dashboard

Q7: Top-Rated Movies and Genres

The treemaps highlight the top 10 individual movies and top genre groups by average rating. Film-Noir based, Mystery based, and Documentary-based genre groups lead in average rating, while War based and Western based also perform strongly — notable given these are lower-volume genre groups (per Q4), suggesting smaller, niche classic-film categories can carry disproportionately high viewer satisfaction and may be worth featuring prominently in the Movie Talkies: Classic campaign.

Q8: Age-Rating Relationship and Rating Trends Over Time

The scatter plot shows a positive relationship between user age group and average rating: older age groups (55+) rate movies consistently higher (≈ 3.8) than younger users (25-34, ≈ 3.5), indicating that older viewers on the platform tend to be more generous or more discerning raters of classic content. The year-over-year rating trend line shows average ratings holding relatively steady across most of the classic era, with a gentle downward drift moving toward 2000.

8. Key Insights and Recommendations

- Comedy and Drama drive the highest engagement volume, but niche genre groups (Film-Noir, Mystery, Documentary, War, Western) achieve the highest average

ratings , a strong candidate set for featured “hidden gem” promotion in the campaign.

- The user base skews male (71.71%) and is concentrated in the 25-34 age group; campaign messaging aimed at female viewers and older age segments could help broaden the platform's audience, aligning with the stated goal of attracting newer customers.
- Older users (55+) consistently rate movies more favorably than younger users, suggesting classic content resonates most strongly with this segment — useful for prioritizing retention versus acquisition messaging by age group.
- Movie catalog depth is heavily weighted toward Drama and Comedy; genres with high average ratings but low catalog volume (e.g., Western, Musical, Documentary) represent an opportunity for catalog expansion.
- User geography, approximated via zip-code region, shows the West/Pacific and Midwest as the largest user bases relevant for any regionally targeted marketing efforts.

9. Conclusion

This analysis of the MovieLens classic movie catalog (releases through 2000) surfaces clear patterns in genre popularity versus genre quality, demographic skew across gender and age, and regional user distribution. These findings directly support the Movie Talkies: Classic campaign's dual goals of deepening engagement with existing high-preference viewers and attracting new audiences — particularly by spotlighting highly-rated niche genres and tailoring outreach toward underrepresented demographic segments such as younger and female viewers.